

MBA THOMAS

AI/ML Engineer | Backend Engineer

FCT Abuja, Nigeria | +2348125116911 | mba.thomas08@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Backend Engineer with 4+ years of experience building intelligent applications, machine learning systems, and production-ready backend services. Experienced in developing predictive models and Generative AI solutions using Python, PyTorch, Scikit-learn, LLMs, RAG, vector embeddings, and AI agents. Skilled in building RESTful APIs with FastAPI, designing PostgreSQL-backed systems, integrating third-party APIs, automating business workflows and deploying containerized applications with Docker and AWS.

SKILLS

- **Programming:** Python, JavaScript, Java, PHP, SQL
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Feature Engineering, Predictive Modeling, Computer Vision
- **Generative AI:** LLMs, OpenAI, RAG, LangChain, LangGraph, MCP, Vector Embeddings, Pinecone, AI Agents
- **Backend Engineering:** FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Alembic, API Integration, OOP
- **AI Automation:** n8n, Webhooks, Workflow Automation, Third-party API Integration
- **Data Science:** Pandas, NumPy, Data Analysis, Data Processing
- **Cloud & DevOps:** AWS, Docker, GitHub Actions, Nginx, Kubernetes

EXPERIENCE

LOCTECH

Nov 2024 - Present

AI/ML & BACKEND ENGINEER

- Designed and developed end-to-end AI applications using Python, PyTorch, vector embeddings, and feature-engineering pipelines, enabling intelligent automation and data-driven decision-making.
- Built scalable RESTful APIs with Python and FastAPI, integrating machine learning models, LLMs, databases, and third-party APIs to deliver AI-driven backend services.
- Developed RAG and vector-search systems using document processing, embeddings, vector databases, and LLMs to enable context-aware question answering over domain-specific knowledge.
- Designed and automated business workflows using n8n, AI agents, webhooks, and third-party API integrations, reducing repetitive manual processes and improving operational efficiency.
- Designed backend architectures using FastAPI, PostgreSQL, SQLAlchemy, and Alembic, and containerized and deployed AI applications using Docker, AWS, and Nginx.

ADALTech

Sep 2023 - Oct 2024

AI & MLOps ENGINEER

- Designed and deployed AI-powered applications incorporating LLMs, RAG pipelines, AI agents, vector embeddings and external AI APIs.
- Built and deployed Python-based AI services using FastAPI, Docker, Uvicorn and AWS, integrating backend APIs with AI/ML workflows.
- Developed automated AI workflows and backend services integrating LLMs, databases, APIs and asynchronous processing to support intelligent business applications.

Orinoco Global

Oct 2022 - Aug 2023

SITE RELIABILITY ENGINEER(SRE)

- Built and maintained automated CI/CD pipelines using GitHub Actions, integrating continuous delivery workflows into product development.
- Optimized system performance, proactively resolved incidents, and improved platform reliability and service availability.
- Spearheaded Infrastructure as Code (IaC) practices for consistent environment provisioning and reduced configuration drift.
- Deployed and managed containerized services using Docker and Kubernetes, supporting application scalability and resilience.

Orinoco Global

Sep 2021 - Oct 2022

FULL STACK PHP DEVELOPER

- Led frontend and backend development using Laravel, PHP, JavaScript, and MySQL, building web applications and backend services from the ground up.

- Designed and maintained relational databases using MySQL and Laravel migrations, supporting data integrity and application reporting.
- Collaborated with engineers, designers, and project managers using Git and Agile development practices to deliver and maintain production web applications.

SELECTED PROJECTS

KNOWLEDGEHUB – ENTERPRISE AI KNOWLEDGE PLATFORM

- Designed and developed a multi-tenant AI knowledge platform using Python, FastAPI, PostgreSQL, SQLAlchemy and Alembic for organizations to manage and query internal knowledge.
- Implemented Retrieval-Augmented Generation (RAG) using document processing, vector embeddings, semantic search, vector databases, and LLMs to provide context-aware responses.
- Built backend architecture supporting authentication, organization and membership management, data persistence, and AI-powered workflows for scalable enterprise AI applications.

ENTERPRISE RAG & SEMANTIC SEARCH SYSTEM

- Developed an end-to-end RAG pipeline for intelligent question answering over domain-specific documents.
- Implemented document ingestion, text chunking, vector embedding generation, Pinecone indexing and semantic similarity search for efficient knowledge retrieval.
- Integrated retrieved context with LLMs to generate relevant responses and optimized retrieval workflows to improve response relevance.

TREND INTELLIGENCE API

- Designed and developed a Python/FastAPI REST API for processing and exposing data-driven intelligence capabilities to client applications.
- Containerized the backend using Docker, creating a reproducible application environment for development and deployment.
- Implemented API-oriented architecture for integrating data-processing functionality with client applications and external services.

SOCCER PREDICTION MODEL

- Developed a PyTorch-based machine learning prediction system for soccer match outcomes using historical match, team-performance, and player-related data.
- Engineered predictive features from historical results, team form, performance statistics, and match-related data to support automated inference and model retraining.
- Containerized the prediction system using Docker Compose and automated fixture processing, prediction generation, CSV reporting and Telegram notifications.

AI VIDEO GENERATION AUTOMATION

- Designed an automated Generative AI pipeline using n8n, LLMs, image-generation APIs, and video-generation services to transform content ideas into video assets.
- Orchestrated script generation, image generation, video creation and media processing through interconnected APIs and workflow automation.
- Implemented API integration and error-handling workflows to manage external-service responses and automate the end-to-end content-generation process.

EDUCATION

AI Technologies <i>AI & Machine Learning Engineering</i> • Achievements: Degree Awarded with Professional Diploma in AI & Machine Learning Engineering	July 2026 <i>Abuja, Nigeria</i>
Cavendish University Uganda <i>Higher Education Certificate, Humanities</i> • Achievements: Degree Awarded with Second Class Upper	May 2024 <i>Kampala, Uganda</i>
NIIT <i>Diploma, Information Technology (Software Engineering)</i> • Achievements: Degree Awarded with Professional Diploma in I.T.	Mar 2023 <i>Lagos, Nigeria</i>
University of Jos <i>B.Sc., Computer Science</i> • Achievements: Degree Awarded with Second Class Honors, Upper Division	Aug 2021 <i>Jos, Nigeria</i>